

Adversarial Quiz Rationale – Otsu’s Thresholding

This quiz is designed to test deep conceptual understanding of Otsu’s thresholding by exposing common misconceptions, including confusion between global and adaptive methods, misunderstanding histogram distributions, and misinterpreting variance optimisation.

Q1 – Objective of Otsu

Correct: B. Otsu minimises within-class variance.

Distractors:

A reflects a common misconception that thresholding improves brightness.

C confuses thresholding with edge detection.

D assumes thresholding removes noise, which is incorrect.

Q2 – Histogram Shape

Correct: B. Works best for bimodal distributions.

Distractors:

A and C ignore the need for separable classes.

D reflects the misconception that skewed data is sufficient.

Q3 – Unimodal Failure

Correct: B. No clear class separation.

Distractors:

A and C are unrelated.

D confuses colour diversity with histogram structure.

Q4 – Key Assumption

Correct: B. Two-class separation.

Distractors:

A is secondary, not core.

C assumes a noise model unnecessarily.

D is irrelevant.

Q5 – Uneven Illumination

Correct: B. Global threshold fails.

Distractors:

A reverses expected behaviour.

C and D are unrelated effects.

Q6 – Alternative Method

Correct: B. Adaptive thresholding.

Distractors:

A and C are preprocessing, not solutions.
D is unrelated.

Q7 – Global Thresholding

Correct: B. One threshold for entire image.

Distractors:

A and C describe adaptive methods.

D is incorrect.

Q8 – Variance Formulation

Correct: B. Maximises between-class variance.

Distractors:

A, C, D misunderstand optimisation objective.

Q9 – Noise Effect

Correct: C. Distorts histogram.

Distractors:

A ignores impact.

B and D reflect incorrect assumptions.

Q10 – Failure Case

Correct: C. Multiple intensity regions.

Distractors:

A and B are ideal conditions.

D is misleading.

Overall, this quiz challenges intuitive but incorrect assumptions, ensuring that learners demonstrate true conceptual understanding rather than memorisation.